UNIVERSITÉ GRENOBLE ALPES

Numerical and experimental study of the thermo-hydro-mechanical behavior of concrete under severe accident conditions using a poro-mechanical approach

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Abstract

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This thesis investigates the thermo-hydro-mechanical behavior of concrete under severe accident conditions using a poro-mechanical approach.

The study combines numerical modeling and experimental validation to better understand the coupled physics involved.

Nomenclature

Finite Element notation

$[C]$	Capacity matrix	
$[K]$	Global stiffness matrix	
$\{F\}$	Nodal force vector	
$\{R\}$	Residual vector	
$\{U\}$	Nodal displacement vector	

Hydraulic symbols

ε_{rd}	Desiccation shrinkage strain related to drying	[-]
ε_{re}	Endogenous shrinkage strain related to hydration	[-]

Mechanical symbols

$\boldsymbol{\sigma}$	Cauchy stress tensor	[Pa]
$\boldsymbol{\varepsilon}$	Total strain tensor	[-]
$\boldsymbol{\varepsilon}_e$	Elastic strain tensor	[-]
$\boldsymbol{\varepsilon}_{th}$	Thermal strain	[-]
$\mathbf{C}$	Fourth-order isotropic elasticity tensor defined by E and ν	[-]
D	Damage variable	[-]
E	Young's modulus	[Pa]
f_t	Tensile strength	[Pa]
G_f	Fracture energy	[J·m^{-2}]
L_e	Characteristic element length	[m]

Thermal symbols

λ	Thermal conductivity	$[\mathrm{W \cdot m^{-1} \cdot K^{-1}}]$
ρ	Mass density	$[\mathrm{kg \cdot m^{-3}}]$
σ	Stefan–Boltzmann constant	$[\mathrm{W \cdot m^{-2} \cdot K^{-4}}]$
ε	Emissivity	[-]
c_p	Specific heat capacity	$[\mathrm{J \cdot kg^{-1} \cdot K^{-1}}]$
h	Convective heat transfer coefficient	$[\mathrm{W \cdot m^{-2} \cdot K^{-1}}]$
q''	Heat flux	$[\mathrm{W \cdot m^{-2}}]$
T	Temperature	[K]

Contents

Chapter 1

Introduction

[REVIEW] This chapter is currently a draft version and will undergo further review and revision.

1.1 Context and problem statement

Concrete containment structures under severe-accident thermal loading present coupled thermo-hydro-mechanical challenges [3].

1.1.1 Concrete behaviour under high temperature

Main contextes:

- Concrete in nuclear containment structures is exposed to extreme temperatures during severe accidents.
- Rapid heating leads to strong thermal stresses, dehydration, cracking and possible spalling govern structural integrity.

1.1.2 Problem statement

Main problems:

- Complex multi-physics coupling under high temperature.
- Temperature- and damage-dependent evolution of transport properties with cracking and stiffness degradation altering the permeability and pore pressure.

- Reliable prediction requires a consistent thermo-hydro-mechanical (THM) framework.

1.1.3 Limitations of existing THM models

Main limitations of existing THM models:

- Lack of fully coupled interactions between the damage state and transport properties.
- Lack of consistent regularization for strain localization.
- Limited probabilistic assessment of parameter uncertainty and sensitivity.

1.2 Objectives and scope of the thesis

Main objectives of the thesis:

- Understanding and Developing a THM model
- Temperature- and damage-dependent evolution of stiffness and transport properties.
- Numerical implementation and numerical validation.
- Probabilistic sensitivity analysis: Evaluating parameter uncertainty and identifying the dominant physical parameters.

The scope of this work is limited to:

- Thermal boundaries limited by severe accident conditions.
- Continuum-scale THM modelling by Cast3M.
- Damage modelling including temperature-dependent stiffness and transport properties.
- Probabilistic sensitivity analysis at the material parameter level.

1.3 Research enviroment and Computational tools

1.3.1 3SR Laboratory

The host institution providing the scientific environment, experimental data, and expertise in structural mechanics and risk analysis.

1.3.2 Finite element code - Cast3M

Cast3M (or Castem) is an open-source finite-element code developed at CEA, which is used to implement advanced constitutive laws and custom multiphysics formulations.

Compared with COMSOL Multiphysics, Cast3m typically offers greater transparency and control for implementing tightly coupled THM while COMSOL offers a user-friendly GUI for many built-in multiphysics interfaces.

1.3.3 Probabilistic analysis tool - Persalys

Persalys is the computational tool dedicated to managing parameter uncertainty in material properties.

It is used to perform the probabilistic sensitivity analysis, generate random sampling (e.g., Monte Carlo simulations), and rank the dominant THM parameters.

Chapter 2

Theoretical and Numerical framework

[REVIEW] This chapter is currently a draft version and will undergo further review and revision.

2.1 Thermal behavior

2.1.1 Conduction, convection and radiation

Heat transfer is the transport of thermal energy driven by a temperature difference. The following definitions of heat transfer modes are summarized from standard references [2].

Conduction

Conduction is the transfer of energy from the more energetic to the less energetic particles of a substance

- **Physical Mechanism:**
- **Rate Equation (Fourier's Law):** For a one-dimensional plane wall having a temperature distribution $T(x)$, the rate equation is expressed as Fourier's law:

$$q''_{cond} = -\lambda \frac{dT}{dx} \tag{2.1}$$

  Where q''_{cond} is the heat flux (W/m^2), λ is the thermal conductivity ($W/m \cdot K$), and the minus sign indicates that heat is transferred in the direction of decreasing temperature.

Convection

The convection heat transfer mode is

- **Physical Mechanism:** Convection heat transfer occurs between a fluid in motion and a bounding surface when the two are at different temperatures
- **Rate Equation (Newton's Law of Cooling):** Regardless of the nature of the convection process, the appropriate rate equation is:

$$q''_{conv} = h(T_s - T_\infty) \tag{2.2}$$

Where q'' is the convective heat flux (W/m^2), h is the convection heat transfer coefficient $(W/m^2 \cdot K)$, T_s is the surface temperature, and T_∞ is the fluid temperature.

Radiation

Thermal radiation is energy emitted by matter that is at a nonzero temperature, and unlike conduction and convection, it does not require the presence of a material medium.

- **Physical Mechanism:** The energy of the radiation field is transported by electromagnetic waves (or photons)
- **Emissive Power (Stefan-Boltzmann law):** The upper limit to the emissive power (rate of energy released per unit area) for an ideal radiator or blackbody is given:

$$E_b = \sigma T_s^4 \tag{2.3}$$

The heat flux emitted by a real surface is given by $E = \varepsilon\sigma T_s^4$, where σ is the Stefan-Boltzmann constant and ε is the emissivity.

- **Net Radiation Exchange:** For a gray surface at T_s completely surrounded by a much larger isothermal surface at T_{sur}, the net rate of radiation heat transfer per unit area is:

$$q''_{rad} = \varepsilon\sigma(T_s^4 - T_{sur}^4) \tag{2.4}$$

2.1.2 Heat transfer

Heat Equation

Based on the principle of energy conservation and Fourier's law, the general heat diffusion equation (or heat equation) for a medium is detailed in [4] and can be expressed as:

$$\rho c_p \frac{\partial T}{\partial t} + \mathrm{div}(-\lambda \overrightarrow{\mathrm{grad}}(T)) - q = 0 \tag{2.5}$$

Where:

- ρ is the volumetric mass density (kg/m^3).
- c_p is the specific heat capacity $(J/kg \cdot K)$.
- λ is the thermal conductivity $(W/m \cdot K)$.
- T is the temperature field and t is time.
- q is the volume heat flux source (W/m^3).

To solve this equation, appropriate boundary conditions must be specified:

- **Prescribed temperatures :** The temperature is specified as $T = T_{imp}$ on the boundary surface ∂V^T.
- **Prescribed surface heat flux :** The total heat flux due to conduction, convection, and radiation on the boundary surface ∂V^φ is given by:

  $$\vec{n} \cdot (\lambda \overrightarrow{\mathrm{grad}}(T)) = \varphi_{imp} + h(T_f - T) + \varepsilon\sigma(T_\infty^4 - T^4) \tag{2.6}$$

  where φ_{imp} is the prescribed heat flux, $h(T_f - T)$ represents the convection term, and $\varepsilon\sigma(T_\infty^4 - T^4)$ is the radiation term.

Thermal properties

The thermal properties of concrete are strongly influenced by both the hydric state and the material composition.

The overall heat capacity ρC_p reflects the multiphase nature of the porous medium, combining contributions from the solid skeleton, liquid water, dry air, and water vapor:

$$\rho C_p = (1 - \phi)\, \rho_s\, C_{ps} + \phi\, [S_l\, \rho_l\, C_{pl} + (1 - S_l)\, (\rho_a\, C_{pa} + \rho_v\, C_{pv})] \tag{2.7}$$

The thermal conductivity depends on saturation, porosity, and temperature:

$$\lambda = \lambda_d \left(1 + 4\, \frac{S_l\, \phi\, \rho_l}{(1 - \phi)\, \rho_s}\right) \tag{2.8}$$

with:

$$\lambda_d = \lambda_{d0}\, [1 + A_\lambda\, (T - T_0)] \tag{2.9}$$

As drying progresses ($S_l \to 0$), both ρC_p and λ decrease, which accelerates the penetration of the thermal front into the material. These dependencies highlight the influence of the hydric state on the thermal field, which is further discussed in Section 2.4.1.

Discretization

Using the FE discretization, the temperature and temperature gradient in the spatial domain are expressed in matrix form as:

$$T(x) = [N(x)]\{T\}, \quad \overrightarrow{\mathrm{grad}}(T) = [B(x)]\{T\} \tag{2.10}$$

From the weak formulation of the heat equation combined with the spatial discretization, we obtain the general matrix equation:

$$[C]\{\dot{T}\} + [K]\{T\} = \{F\} \tag{2.11}$$

Where the matrices are defined as follows:

- **Capacity matrix:**
$$[C] = \int_V \rho c_p [N]^T [N] dV \quad (J/K) \tag{2.12}$$
- **Conductivity matrix:**
$$[K] = \int_V [B]^T [\lambda][B] dV + \int_{\partial V^\varphi} h [N]^T [N] dS \quad (W/K) \tag{2.13}$$
- **Equivalent nodal flux vectors:**
$$\{F\} = \int_V [N]^T q dV + \int_{\partial V^\varphi} [N]^T \left(\varphi_{imp} + hT_f + \varepsilon\sigma(T_\infty^4 - T^4) \right) dS \quad (W) \tag{2.14}$$

For steady-state thermal analysis, the temperature is time-independent ($\{\dot{T}\} = 0$) (assuming constant material properties and neglecting radiation):

$$[K]\{T\} = \{F\} \tag{2.15}$$

By solving this linear system, we can determine the unknown temperatures $\{T\}$ at the nodes across the entire global mesh.

2.2 Hydric behavior

[REVIEW]

When concrete is exposed to high temperatures, the behavior of water within the porous network plays a central role in the thermo-hydro-mechanical response of the material. This section presents the classification of water in concrete, the phase-change mechanisms, the governing equation for moisture transport, and the constitutive laws required to close the system.

2.2.1 Types of water in concrete

Concrete is a multiphase porous material whose voids can be filled with water, air and other fluids. The water present in the cement paste can be broadly classified into two categories [3]

- **Pore water (free/evaporable water):** This water evaporates when the thermodynamic conditions are met (*i.e.*, at $T = 100\,°\mathrm{C}$ and atmospheric pressure) and exists in three main forms:
  - *Capillary and inter-hydrate water:* fills the capillary pores (size $\sim$ 100s of nm) and inter-hydrate pores(size $\sim$ 10s of nm), corresponding to a saturation $S_{ssp} < S_l < 1$.
  - *Physically adsorbed water (gel water):* retained by the surface of the solid matrix due to Van der Waals forces, confined in gel pores (size $\sim$ a few nm).
  - *Interlayer water:* confined in $\sim$ 1 nm space between the C-S-H layers.
- **Chemically bound water (non-evaporable):** This water is combined with anhydrous cement during hydration and forms part of the crystalline structure of hydration products (e.g., C-S-H, portlandite). It can only be released through thermal decomposition (dehydration) at elevated temperatures.

It is important to note that the range of pore sizes is a continuous one; the dimensional boundaries between capillary, inter-hydrate and gel pores are approximations to inherently blurred boundaries.

2.2.2 Evaporation and dehydration

Two distinct phase-change mechanisms govern the release of water in heated concrete:

- **Evaporation** is the transition of free liquid water into vapor. It occurs progressively as temperature increases, becoming significant above approximately 100 °C at atmospheric pressure. The evaporation rate $\dot{m}_{vap}$ is determined from the mass conservation of liquid water (see Section 2.2.3).

- **Dehydration** is the thermally-driven decomposition of hydration products, releasing chemically bound water as vapor. The main reactions occur at characteristic temperature ranges [3]:
  - Up to 300 °C: progressive decomposition of C-S-H phases and ettringite;
  - 400–500 °C: dehydroxylation of portlandite ($Ca(OH)_2 \rightarrow CaO + H_2O$);
  - $\sim$ 570 °C: α–β quartz transformation;
  - Above 600 °C: decarbonation of calcareous aggregates ($CaCO_3 \rightarrow CaO + CO_2$).

Both evaporation and dehydration act as vapor sources that increase gas pressure within the pore network. They also consume latent energy, affecting the temperature field (see Section 2.4).

The cumulative dehydrated water mass is described by the following phenomenological relationship

$$\Delta m_{dehyd} = f_s \cdot f_m \cdot f_c \cdot f(T) \tag{2.16}$$

where f_s is a stoichiometric factor, f_m accounts for the concrete's age, f_c is the cement mass per unit volume, and $f(T)$ is a temperature-dependent function:

$$f(T) = \begin{cases} 0 & \text{if } T < 105\,^\circ\text{C} \\ \left[1 + \sin\left(\frac{\pi}{2}\left(1 - 2\exp(-0.004\,(T - 105))\right)\right)\right] & \text{if } T \geq 105\,^\circ\text{C} \end{cases} \tag{2.17}$$

2.2.3 Moisture transport equation

Transport mechanisms

The transport of moisture in heated concrete is governed by two mechanisms:

- **Advective flow (Darcy's law):** The bulk flow of liquid water and gas through the porous network is driven by pressure gradients:

$$\mathbf{v}_{l-s} = -\frac{K\,k_{rl}}{\mu_l}\,(\nabla p_g - \nabla p_c) \tag{2.18}$$

$$\mathbf{v}_{g-s} = -\frac{K\,k_{rg}}{\mu_g}\nabla p_g \tag{2.19}$$

where K is the intrinsic permeability, k_{rl} and k_{rg} are the relative permeabilities of liquid and gas, μ_l and μ_g are the dynamic viscosities, and $p_l = p_g - p_c$ is the liquid pressure.

- **Diffusive flow (Fick's law):** Water vapor diffuses within the gas mixture according to:
$$\mathbf{j}_v = -\rho_g \, D \, \nabla \left(\frac{\rho_v}{\rho_g} \right) \tag{2.20}$$
where D is the effective diffusivity accounting for the tortuous porous network.

Mass conservation equations

The mathematical framework for moisture transport in heated concrete is based on the conservation of mass for each phase. The general form of a mass conservation equation for a phase π is [3]:
$$\frac{\partial}{\partial t}(\text{density}) + \nabla \cdot (\text{flux}) = \text{source} \tag{2.21}$$

Concrete at high temperature is treated as a multiphase porous medium consisting of a solid skeleton (s), liquid water (l), water vapor (v), and dry air (a). Applying Equation (2.21) to each phase yields [3]:

- **Solid skeleton:**
$$\frac{\partial m_s}{\partial t} = \dot{m}_{dehyd} \tag{2.22}$$
- **Liquid water:**
$$\frac{\partial m_l}{\partial t} + \nabla \cdot (m_l \, \mathbf{v}_{l-s}) = -\dot{m}_{vap} - \dot{m}_{dehyd} \tag{2.23}$$
- **Water vapor:**
$$\frac{\partial m_v}{\partial t} + \nabla \cdot (m_v \, \mathbf{v}_{g-s}) + \nabla \cdot (m_v \, \mathbf{v}_{v-g}) = \dot{m}_{vap} \tag{2.24}$$
- **Dry air:**
$$\frac{\partial m_a}{\partial t} + \nabla \cdot (m_a \, \mathbf{v}_{g-s}) + \nabla \cdot (m_a \, \mathbf{v}_{a-g}) = 0 \tag{2.25}$$

where $\mathbf{v}_{\pi-s}$ is the velocity of phase π ($\pi = l, g$) relative to the solid skeleton, and $\mathbf{v}_{\pi-g}$ is the velocity of species π ($\pi = v, a$) relative to the gas mixture. The source term $\dot{m}_{vap}$ represents the evaporation rate ($\dot{m}_{vap} > 0$ for evaporation, < 0 for condensation), and $\dot{m}_{dehyd}$ is the dehydration rate ($\dot{m}_{dehyd} < 0$, treated as a mass loss of the solid skeleton and a mass source for liquid water).

The mass of each phase per unit volume of porous medium is [3]:

$$m_s = (1-\phi)\,\rho_s \tag{2.26}$$

$$m_l = \rho_l\, S_l\, \phi \tag{2.27}$$

$$m_v = \rho_v\,(1-S_l)\,\phi \tag{2.28}$$

$$m_a = \rho_a\,(1-S_l)\,\phi \tag{2.29}$$

where ϕ is the porosity and S_l is the liquid saturation degree.

By combining the mass conservation of liquid water and water vapor, the evaporation source term $\dot{m}_{vap}$ is eliminated, yielding the **water species conservation equation**:

$$\begin{aligned} &S_l\rho_l\frac{\partial\phi}{\partial t} + S_l\phi\frac{\partial\rho_l}{\partial t} + \rho_l\phi\frac{\partial S_l}{\partial t} + (1-S_l)\phi\frac{\partial\rho_v}{\partial t} - \rho_v\phi\frac{\partial S_l}{\partial t} + (1-S_l)\rho_v\frac{\partial\phi}{\partial t} \\ &+ \nabla\cdot\left(-K\frac{\rho_l k_{rl}}{\mu_l}\nabla p_l\right) + \nabla\cdot\left(-K\frac{\rho_v k_{rg}}{\mu_g}\nabla p_g - D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_v}{p_g}\right) = -\dot{m}_{dehyd} \end{aligned} \tag{2.30}$$

This equation has three primary unknowns: the temperature T, the capillary pressure p_c, and the gas pressure p_g. Similarly, the **dry air conservation equation** reads:

$$(1-S_l)\phi\frac{\partial\rho_a}{\partial t} - \rho_a\phi\frac{\partial S_l}{\partial t} + (1-S_l)\rho_a\frac{\partial\phi}{\partial t} + \nabla\cdot\left(-K\frac{\rho_a k_{rg}}{\mu_g}\nabla p_g - D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_a}{p_g}\right) = 0 \tag{2.31}$$

Constitutive relations for the gas phase

The gaseous phase (dry air + water vapor) is governed by the following relations [3]:

- **Ideal gas law:**

$$\rho_\pi = \frac{p_\pi M_\pi}{RT}, \quad \pi = a, v \tag{2.32}$$

  where M_π is the molar mass and R is the universal gas constant.

- **Dalton's law:**

$$p_g = p_a + p_v \tag{2.33}$$

- **Kelvin equation** (thermodynamic equilibrium between liquid and vapor):

$$p_v = p_{vs}(T)\,\exp\left(\frac{M_v}{\rho_l RT}(p_g - p_c - p_{vs})\right) \tag{2.34}$$

  where $p_{vs}(T)$ is the saturation vapor pressure, approximated by the Hyland-Wexler relationship:

$$p_{vs}(T) = \exp\left[\frac{C_1}{T} + C_2 + C_3T + C_4T^2 + C_5T^3 + C_6\ln(T)\right] \tag{2.35}$$

Sorption isotherms

The relationship between capillary pressure and liquid saturation is described by the Van Genuchten model:

$$p_c = a\left(S_l^{-b} - 1\right)^{1-1/b} \tag{2.36}$$

where a and b are material parameters determined experimentally. Representative values at ambient temperature are given in Table 2.1.

Table 2.1: Van Genuchten parameters at ambient temperature

	NSC	HPC	NSM	HPM
a [MPa]	18.62	46.94	37.55	96.28
b [-]	2.275	2.060	2.168	1.954

At elevated temperatures, the parameter a is modified to account for changes in surface tension and pore structure:

$$S_l = \left(\left(\frac{E(T)}{a(T)}\,p_c\right)^{b/(b-1)} + 1\right)^{-1/b} \tag{2.37}$$

where $E(T)$ accounts for the surface tension evolution and $a(T)$ accounts for microstructure changes above 100 °C.

Intrinsic permeability

Permeability is a key parameter governing moisture transport. For a thermo-hydral problem, the following relationship is adopted:

$$K = K_0 \cdot 10^{A_T(T-T_0)}\left(\frac{p_g}{p_{atm}}\right)^{A_p} \tag{2.38}$$

where K_0 is the intrinsic permeability at reference conditions, $A_T = 0.005$ and $A_p = 0.368$.

When mechanical damage is considered, the permeability increases significantly:

$$K = K_0 \cdot 10^{A_T(T-T_0)}\left(\frac{p_g}{p_{atm}}\right)^{A_p} \cdot 10^{A_D\,\mathcal{D}} \tag{2.39}$$

where $\mathcal{D}$ is the damage parameter and A_D is a material constant. This damage-permeability coupling is a key mechanism in the THM model (see Section 2.4).

Relative permeability

The relative permeabilities of liquid and gas are described by the Van Genuchten relationships:

$$k_{rl} = \sqrt{S_l}\left(1 - \left(1 - S_l^{1/A_S}\right)^{A_S}\right)^2 \tag{2.40}$$

$$k_{rg} = \sqrt{1 - S_l}\left(1 - S_l^{1/A_S}\right)^{2A_S} \tag{2.41}$$

where $A_S = 0.6$.

Effective diffusivity

The diffusivity of vapor in air at a given temperature and pressure is:

$$D_{va}(T, p_g) = D_{v0}\left(\frac{T}{T_0}\right)^{B_v}\frac{p_{g0}}{p_g} \tag{2.42}$$

where $D_{v0} = 2.58 \times 10^{-5}$ m^2/s and $B_v = 1.667$. The effective diffusivity in the porous medium accounts for saturation and tortuosity:

$$D = (1 - S_l)^{A_v}\,\phi\,\tau\,D_{va}(T, p_g) \tag{2.43}$$

with the tortuosity:

$$\tau(\phi, S_l) = \phi^{1/3}(1 - S_l)^{7/3} \tag{2.44}$$

Porosity evolution

The porosity increases with temperature due to microstructural changes:

$$\phi = \phi_0 + A_\phi\,(T - T_0) \tag{2.45}$$

where ϕ_0 is the initial porosity and A_ϕ is a material constant.

2.2.4 Pore pressure build-up

When concrete is heated, the evaporation of free water and the release of chemically bound water (dehydration) produce vapor, raising the pressure in the pores. Due to pressure gradients, part of this vapor is evacuated toward the heated face, while another part is driven toward the interior, cooler regions of the specimen. Once the vapor reaches zones where the temperature is sufficiently low, it condenses back into liquid water. This process continues until a quasi-saturated zone is formed, known as the *moisture clog*. This saturated layer acts as an impermeable barrier that blocks further vapor migration toward the colder region, giving rise to significant pore pressure build-up behind the drying front. In dense materials such as high-performance concrete (HPC), the low intrinsic permeability (K_0) further aggravates this phenomenon by limiting the overall gas transport capacity.

The magnitude of pore pressure depends on:

- the rate of vapor production (controlled by the heating rate and initial moisture content);
- the intrinsic permeability K_0 and its evolution with temperature and damage;
- the porosity ϕ, which governs the available volume for gas storage.

When the pore pressure exceeds the tensile strength of the material, it can trigger cracking and potentially explosive spalling. This mechanism is discussed in more detail in Section 2.5.2.

2.3 Mechanical behavior

[REVIEW]

The mechanical response of concrete under high-temperature conditions is characterized by an initial linear elastic phase, followed by severe stiffness degradation due to the formation of micro-cracks. In this numerical framework, the behavior is modeled by coupling a linear elastic constitutive law with a continuum damage mechanics approach.

2.3.1 Linear elastic model

In the present work, concrete is modeled as a linear elastic material. The effective stress tensor $\tilde{\boldsymbol{\sigma}}$, defined as the stress acting on the undamaged material, is related to the elastic strain tensor $\boldsymbol{\varepsilon}_e$ through the isotropic Hooke's law:

$$\tilde{\boldsymbol{\sigma}} = \mathbf{C} : \boldsymbol{\varepsilon}_e \tag{2.46}$$

where $\mathbf{C}$ is the constant fourth-order elastic stiffness tensor, defined by the Young's modulus E and Poisson's ratio ν.

The elastic strain is obtained from the total strain by subtracting the free (stress-free) strain components:

$$\boldsymbol{\varepsilon}_e = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{th} - \boldsymbol{\varepsilon}_{sh} \tag{2.47}$$

where the thermal strain $\boldsymbol{\varepsilon}_{th}$ and the shrinkage strain $\boldsymbol{\varepsilon}_{sh}$ are defined in Section 2.4.2.

Linear finite element formulation

[REVIEW]

[REVIEW] Further references are needed to strengthen the scientific basis of this section. Such as "Numerical Methods for Nonlinear Mechanics lectures" by Stefano DAL PONT

Applying the principle of virtual work, the global linear system takes the well-known fundamental form:

$$[\mathbf{K}]\{\mathbf{U}\} = \{\mathbf{F}\} \tag{2.48}$$

where $[\mathbf{K}] = \int_V \mathbf{B}^T \mathbf{C} \mathbf{B} dV$ is the global elastic stiffness matrix, and $\{\mathbf{F}\}$ is the vector of equivalent nodal forces In Cast3M, this linear system is directly assembled and solved using the `RIGI` and `RESO` operators.

2.3.2 Mazars damage model

Concrete cracks when tensile strains become too large. In this work, this behaviour is represented with the Mazars damage model [5].

Damage parameter and equivalent strain

The model introduces a single scalar internal variable $\mathcal{D}$, damage parameter, ranging from 0 (intact material) to 1 (fully broken material). The apparent macroscopic stress is reduced according to the effective stress concept:

$$\boldsymbol{\sigma} = (1 - \mathcal{D})\, \tilde{\boldsymbol{\sigma}} \tag{2.49}$$

The term *effective stress* here refers to the Mazars concept (stress in the undamaged skeleton), and should not be confused with the Biot effective stress $\boldsymbol{\sigma}''$ used in the THM framework (Section 2.4), which accounts for pore pressure effects.

[REVIEW]

Because concrete cracking is primarily governed by tensile extensions, Mazars introduced an *equivalent strain* (ε_{eq}) to translate any 3D multiaxial strain state into a scalar uniaxial indicator. In the case of a thermo-mechanical loading, it is important to note that only the elastic strain tensor $\boldsymbol{\varepsilon}_e$ contributes to the evolution of damage. The equivalent strain is calculated solely from the positive principal elastic strains ε_i:

$$\varepsilon_{eq} = \sqrt{\langle \boldsymbol{\varepsilon}_e \rangle_+ : \langle \boldsymbol{\varepsilon}_e \rangle_+} = \sqrt{\sum_{i=1}^{3} \langle \varepsilon_i \rangle_+^2} \tag{2.50}$$

where the Macaulay brackets $\langle x \rangle_+$ denote the positive part of x (i.e., $\langle x \rangle_+ = x$ if $x \geq 0$, and 0 if $x < 0$).

Damage initiates when the equivalent strain exceeds the initial damage threshold, denoted as ε_{d0}.

$$\varepsilon_{d0} = \frac{f_t}{E}. \tag{2.51}$$

At the damage onset ($\varepsilon_{eq} = \varepsilon_{d0}$), the damage variable is zero:

$$\mathcal{D} = 0 \qquad \text{for} \qquad \varepsilon_{eq} \leq \varepsilon_{d0}. \tag{2.52}$$

For uniaxial tension, damage onset corresponds to:

$$\sigma_t = f_t. \tag{2.53}$$

In this simplified interpretation, f_t denotes the uniaxial tensile strength of the concrete and E is Young's modulus.

Damage evolution laws for tension and compression

To accurately represent concrete behavior, the total damage D is formulated as a combination of two independent damage variables: D_t for tension and D_c for compression.

When the damage threshold is exceeded ($\varepsilon_{eq} > \varepsilon_{d0}$), the evolution of these two modes is explicitly governed by exponential laws:

$$\mathcal{D}_t = 1 - \frac{(1 - A_t)\varepsilon_{d0}}{\varepsilon_{eq}} - A_t \exp\left[-B_t(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{2.54}$$

$$\mathcal{D}_c = 1 - \frac{(1 - A_c)\varepsilon_{d0}}{\varepsilon_{eq}} - A_c \exp\left[-B_c(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{2.55}$$

where A_t, B_t, A_c, and B_c are material parameters identified from uniaxial tensile and compressive tests. These parameters modulate the shape of the post-peak softening curves: parameters A define the residual asymptotic stress level, while parameters B control the slope of the stress drop.

The global macroscopic damage $\mathcal{D}$ is then calculated using a linear combination with weighting coefficients α_t and α_c:

$$\mathcal{D} = \alpha_t^\beta \mathcal{D}_t + \alpha_c^\beta \mathcal{D}_c \tag{2.56}$$

The coefficients α_t and α_c depend on the principal stress state, evaluating the respective contributions of tensile and compressive stresses. In pure tension, $\alpha_t = 1$ and $\alpha_c = 0$; conversely, in pure compression, $\alpha_c = 1$ and $\alpha_t = 0$. The parameter β (typically fixed at 1.06) is a shear-correction coefficient introduced to improve the structural response under shear loadings.

Non-linear equilibrium and Imbalance vector (R)

[REVIEW]

[REVIEW] Further references are needed to strengthen the scientific basis of this section. Such as "Numerical Methods for Nonlinear Mechanics lectures" by Stefano DAL PONT

Due to the damage evolution, the stress-strain relationship becomes non-linear. The mechanical equilibrium is rewritten as:

$$[\mathbf{K}]\{\mathbf{U}\} = \{\mathbf{F}\}_{ext} - \int_V \mathbf{B}^T \boldsymbol{\sigma}_{nl} dV \tag{2.57}$$

To solve this, the `PASAPAS` non-linear procedure evaluates the deviation from equilibrium using an iterative Newton-Raphson-type algorithm. It calculates the *imbalance vector* (or residual), $\{\mathbf{R}\}$, which represents the difference between the external applied forces and the internal resisting forces:

$$\{\mathbf{R}\} = \{\mathbf{F}\}_{ext} - \int_V \mathbf{B}^T \boldsymbol{\sigma} dV \tag{2.58}$$

At each iteration i, the algorithm solves the system using the initial elastic stiffness matrix: $[\mathbf{K}]\{\delta\mathbf{U}\}^{i+1} = \{\mathbf{R}\}^i$. The displacements and internal variables (like D) are continuously updated until the norm of the imbalance vector becomes negligible ($\|\mathbf{R}\| < \epsilon \mathbf{F}_{ref}$).

2.3.3 Mesh dependency and fracture energy regularization

A major drawback of local softening models is the pathological mesh dependency. Mathematically, softening causes a loss of ellipticity in the differential equations, leading to strain localization in a band whose width depends entirely on the finite element size.

[REVIEW]

[REVIEW] This section requires additional references to support the statements presented.

To overcome this issue without resorting to complex non-local formulations, a regularization technique based on Hillerborg's specific fracture energy (G_f) is utilized. Physically, the fracture energy G_f represents the energy required to propagate a discrete crack per unit *area* (expressed in J/m^2). However, in the continuum damage mechanics framework of the Finite Element Method, cracking is modeled as a smeared softening phenomenon occurring over the *volume* of the 3D finite element.

To reconcile the areal energy dissipation of a real crack with the volumetric dissipation of the continuum element, a characteristic length of the finite element, L_e, is introduced. In the Cast3M implementation (using the `@LONGEF` procedure), this equivalent length is evaluated based on the element's geometry, calculated as $L_e = V^{1/D}$ where D is the spatial dimension.

The total fracture energy G_f dissipated by a finite element of size L_e is the product of L_e and the total area under the uniaxial stress-strain ($\sigma - \varepsilon$) softening curve (which represents the volumetric dissipated energy density $g_f = G_f/L_e$). This total energy is strictly partitioned into two components: the reversible elastic strain energy stored up to the peak stress, and the energy dissipated strictly during the non-linear damage softening phase:

$$G_f = G_{elastic} + G_{softening} \tag{2.59}$$

Evaluation of the elastic energy ($G_{elastic}$)

[REVIEW]

[REVIEW] This section requires additional references to support the statements presented.

The elastic energy density up to the onset of damage (when the stress reaches the tensile strength f_t at the initial strain threshold $\varepsilon_{d0} = f_t/E_0$) corresponds to the area of the elastic triangle under the curve:

$$g_{elastic} = \frac{1}{2} f_t \varepsilon_{d0} = \frac{f_t^2}{2E_0} \tag{2.60}$$

Multiplying by the characteristic length yields the areal elastic energy: $G_{elastic} = \frac{f_t^2}{2E_0} L_e$. Consequently, the remaining energy strictly available for the damage softening phase is calculated as:

$$G_{softening} = G_f - \frac{f_t^2}{2E_0} L_e \tag{2.61}$$

Analytical evaluation of $G_{softening}$ and the softening parameter B_t

[REVIEW]

[REVIEW] This section requires additional references to support the statements presented.

In the original Mazars formulation, the evolution of damage in pure tension is governed by an exponential law. Assuming a purely brittle failure in tension ($A_t \approx 1$), the post-peak stress-strain relationship can be simplified to:

$$\sigma(\varepsilon) \approx f_t \exp\left[-B_t(\varepsilon - \varepsilon_{d0})\right] \tag{2.62}$$

The volumetric softening energy density ($g_{softening}$) is the integral of this stress-strain curve from the peak strain ε_{d0} to infinity:

$$g_{softening} = \int_{\varepsilon_{d0}}^{\infty} f_t \exp\left[-B_t(\varepsilon - \varepsilon_{d0})\right] d\varepsilon = \frac{f_t}{B_t} \tag{2.63}$$

By converting this volumetric density into areal energy via the characteristic length L_e, we establish the fundamental relationship between the fracture energy and the Mazars tensile parameter:

$$G_{softening} = g_{softening} \cdot L_e = \frac{f_t \cdot L_e}{B_t} \tag{2.64}$$

Dynamic regularization of B_t (B_t^{mod})

To ensure that each finite element dissipates exactly the objective intrinsic energy G_f regardless of its size L_e, the parameter B_t must be dynamically recalculated. By equating the integral of the Mazars softening curve to the available softening energy, we deduce the modified parameter B_t^{mod}:

$$B_t^{mod} = \frac{f_t \cdot L_e}{G_{softening}} = \frac{f_t \cdot L_e}{G_f - \frac{f_t^2}{2E_0} L_e} \tag{2.65}$$

This energetic adjustment guarantees that the macroscopic crack propagation and global energy dissipation remain completely objective and independent of the selected spatial discretization.

2.4 Coupling mechanisms

The thermal, hydric, and mechanical behaviors presented in the previous sections are not independent: they interact through a network of coupling mechanisms. This section identifies and formulates the two main coupling paths that govern the response of concrete exposed to fire.

2.4.1 Thermo-Hydric (TH) coupling mechanisms

The thermal and hydric fields are coupled in both directions: temperature drives phase changes and modifies transport properties, while fluid flow and phase changes affect the energy balance.

Temperature as a driver of hydric phenomena

Temperature influences the hydric field through three main mechanisms:

- **Phase changes:** As temperature rises, evaporation and dehydration release vapor into the pore network (see Section 2.2.2), acting as mass source terms in the water species conservation Equation (2.30).
- **Temperature-dependent transport properties:** Most constitutive parameters governing moisture transport depend explicitly on temperature [3]: intrinsic permeability K increases (Equation (2.38)), porosity ϕ increases (Equation (2.45)), fluid viscosities μ_l, μ_v, μ_a decrease (enhancing advective flow), saturating vapor pressure p_{vs} increases sharply, liquid water density ρ_l decreases, and the effective diffusivity D increases.
- **Sorption isotherms:** The relationship between capillary pressure and liquid saturation is modified at high temperatures through the temperature-dependent parameter $a(T)$ in the Van Genuchten model (Equation (2.37)).

Energy conservation equation

The energy conservation equation governs the temperature field within the porous medium. Similar to the heat equation in Section 2.1.2, it includes a storage term, a conductive flux, and additionally accounts for advective heat transport and phase-change energy terms [3]:

$$\rho C_p \frac{\partial T}{\partial t} + \underbrace{(m_l C_l \mathbf{v}_{l-s} + m_g C_g \mathbf{v}_{g-s}) \cdot \nabla T}_{\text{advection}} + \nabla \cdot (-\lambda \cdot \nabla T) = \underbrace{-H_{vap}\,\dot{m}_{vap} + H_{dehyd}\,\dot{m}_{dehyd}}_{\text{phase-change energy}} \tag{2.66}$$

Compared to the classical heat equation Equation (2.5) $\rho C_p \frac{\partial T}{\partial t} - \nabla \cdot (\lambda \nabla T) = 0$, two additional contributions appear:

- **Advective heat transport:** The bulk flow of liquid water and gas, driven by pressure gradients, transport enthalpy through the material. This term redistributes energy along the flow paths and can be significant when permeability and pressure gradients are large.
- **Phase-change energy (right-hand side):** Concrete does not self-generate heat; these terms represent energy consumed by evaporation ($-H_{vap}\,\dot{m}_{vap}$, heat sink since $\dot{m}_{vap} > 0$) and dehydration ($+H_{dehyd}\,\dot{m}_{dehyd}$, also a heat sink since $\dot{m}_{dehyd} < 0$). Both locally slow the temperature rise.

After replacing the mass fluxes by Darcy's law (Equations (2.18) and (2.19)), the energy conservation takes its final coupled form:

$$\begin{aligned} \rho C_p \frac{\partial T}{\partial t} - K \left(C_l \frac{\rho_l k_{rl}}{\mu_l} (\nabla p_g - \nabla p_c) + C_g \frac{\rho_g k_{rg}}{\mu_g} \nabla p_g \right) \cdot \nabla T \\ -\nabla \cdot (\lambda \cdot \nabla T) = -H_{vap}\,\dot{m}_{vap} + H_{dehyd}\,\dot{m}_{dehyd} \end{aligned} \tag{2.67}$$

where the overall heat capacity ρC_p (Equation (2.7)) and thermal conductivity λ (Equation (2.8)) are defined in Section 2.1.2.

Thus, the thermal and hydric fields are *strongly coupled*: temperature modifies the source terms and transport coefficients in the moisture equation, while moisture content and fluid flow modify the heat capacity, conductivity, and introduce latent heat and advection terms in the energy equation.

2.4.2 Thermo-hydric input to the mechanical field

The mechanical behavior of heated concrete is driven by the thermal and hydric fields. In the framework adopted here, the coupling is *one-way*: the outputs of the TH problem — temperature T, gas pressure p_g, and capillary pressure p_c — serve as input to the mechanical problem. The mechanical field feeds back into the TH equations only through the damage-enhanced permeability; this coupling is discussed in Section 2.5.1.

Strain decomposition

When a mechanically loaded concrete element is exposed to high temperature, the total strain is decomposed as a superposition of components of different origins [3]:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_\sigma + \boldsymbol{\varepsilon}_{th} + \boldsymbol{\varepsilon}_{sh} \tag{2.68}$$

where:

- $\boldsymbol{\varepsilon}_\sigma$ represents the *mechanical strains* induced by external loading (elastic strain, creep strain, etc.);
- $\boldsymbol{\varepsilon}_{th}$ represents the *thermal dilation strain*, driven by the temperature field;
- $\boldsymbol{\varepsilon}_{sh}$ represents the *capillary shrinkage strain*, driven by the capillary pressure field.

Thermal dilation

The thermal dilation strain is isotropic and produces a uniform expansion of the material, which is expressed in incremental form as [3]:

$$d\boldsymbol{\varepsilon}_{th} = \beta_s(T)\, dT\, \mathbf{I} \tag{2.69}$$

where $\mathbf{I}$ is the second-order identity tensor and $\beta_s(T)$ is the thermal dilation coefficient, which for concrete can be approximated as a linear function of temperature:

$$\beta_s(T) = A_{B1}\,(T - T_0) + A_{B0} \tag{2.70}$$

with A_{B0} and A_{B1} being material constants.

The thermal dilation strain is isotropic and produces a uniform expansion of the material. Near the heated surface, the steep thermal gradient creates a strong differential expansion: the hot surface layer tends to expand while the cooler interior restrains it, inducing compressive stresses at the surface and tensile stresses at depth.

Capillary shrinkage

The drying process modifies the pore pressure state, which in turn induces a volumetric deformation of the solid skeleton. The capillary shrinkage strain is expressed as [3]:

$$d\boldsymbol{\varepsilon}_{sh} = \frac{\alpha}{K_T}\,\frac{dp_s}{dt}\,\mathbf{I} \tag{2.71}$$

where α is the Biot coefficient, K_T is the bulk modulus of the drained skeleton, and p_s is the average pore pressure acting on the solid skeleton, defined as:

$$p_s = p_g - S_l\, p_c = S_g\, p_g + S_l\, p_l \tag{2.72}$$

This expression represents the saturation-weighted average of the gas and liquid pressures. Since p_g and p_c are direct outputs of the TH problem, p_s is entirely determined by the TH solution.

As the concrete dries during heating, the capillary pressure increases (the pores desaturate), leading to a contraction (shrinkage) of the solid matrix. This effect is most pronounced in the drying front region where the saturation gradient is steepest.

Effective stress and pore pressure loading

Beyond the free strains ($\boldsymbol{\varepsilon}_{th}$, $\boldsymbol{\varepsilon}_{sh}$), the pore pressure directly modifies the stress state in the solid skeleton through the concept of effective stress, which is defined as:

$$\boldsymbol{\sigma}'' = \boldsymbol{\sigma} + \mathbf{I}\,\alpha\, p_s \tag{2.73}$$

where $\boldsymbol{\sigma}$ is the total stress tensor and p_s is the average pore pressure defined in Equation (2.72).

In heated concrete, gas pressure builds up due to evaporation and dehydration (see Section 2.2.4). This increase in p_g raises p_s, which reduces the effective compressive stress in the skeleton, effectively pushing the material towards tensile failure. When the tensile effective stress exceeds the tensile strength f_t, cracking initiates.

Note that f_t, along with Young's modulus E and compressive strength f_c, are prescribed as empirical functions of temperature to account for the progressive degradation of the cementitious matrix at high temperatures.

2.5 THM model and spalling

This section assembles the governing equations of the thermo-hydro-mechanical (THM) model from the conservation laws and constitutive relations developed in the previous sections. The coupled system is then used to explain how spalling arises from the interaction between thermal, hydric, and mechanical fields.

2.5.1 Full THM governing formulation

The state of concrete at high temperature is described by four primary variables [3]:

$$\mathcal{U} = \{p_g,\ p_c,\ T,\ \mathbf{u}\} \tag{2.74}$$

where p_g is the gas pressure, p_c is the capillary pressure, T is the temperature, and $\mathbf{u}$ is the displacement vector.

The model consists of four conservation equations, each associated with one primary variable. Their final forms, after substitution of Darcy's law, Fick's law, and the constitutive relations, are as follows.

Dry air conservation

$$\begin{aligned}
&\phi(1-S_l)\left(\frac{\partial \rho_a}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial \rho_a}{\partial p_c}\frac{\partial p_c}{\partial t} + \frac{\partial \rho_a}{\partial p_g}\frac{\partial p_g}{\partial t}\right) + (1-S_l)\rho_a\frac{\partial \phi}{\partial T}\frac{\partial T}{\partial t} \\
&- \phi\rho_a\left(\frac{\partial S_l}{\partial p_c}\frac{\partial p_c}{\partial t} + \frac{\partial S_l}{\partial T}\frac{\partial T}{\partial t}\right) - \nabla\cdot\left(\frac{K\rho_a k_{rg}}{\mu_g}\nabla p_g\right) - \nabla\cdot\left(D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_a}{p_g}\right) = 0
\end{aligned} \tag{2.75}$$

Water species conservation

$$\begin{aligned}
&(S_l\rho_l - (1-S_l)\rho_v)\frac{\partial \phi}{\partial T}\frac{\partial T}{\partial t} + S_l\phi\frac{\partial \rho_l}{\partial T}\frac{\partial T}{\partial t} + \phi(\rho_l-\rho_v)\left(\frac{\partial S_l}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial S_l}{\partial p_c}\frac{\partial p_c}{\partial t}\right) \\
&+ (1-S_l)\phi\left(\frac{\partial \rho_v}{\partial T}\frac{\partial T}{\partial t} + \frac{\partial \rho_v}{\partial p_c}\frac{\partial p_c}{\partial t} + \frac{\partial \rho_v}{\partial p_g}\frac{\partial p_g}{\partial t}\right) \\
&- \nabla\cdot\left(\frac{K\rho_l k_{rl}}{\mu_l}(\nabla p_g - \nabla p_c)\right) - \nabla\cdot\left(\frac{K\rho_v k_{rg}}{\mu_g}\nabla p_g\right) - \nabla\cdot\left(D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_v}{p_g}\right) = -\dot{m}_{dehyd}
\end{aligned} \tag{2.76}$$

Energy conservation

$$\begin{aligned}
\rho C_p\frac{\partial T}{\partial t} - K\left(C_l\frac{\rho_l k_{rl}}{\mu_l}(\nabla p_g - \nabla p_c) + C_g\frac{\rho_v k_{rg}}{\mu_g}\nabla p_g\right)\cdot\nabla T - \nabla\cdot(\lambda\cdot\nabla T) \\
= -H_{vap}\,\dot{m}_{vap} + H_{dehyd}\,\dot{m}_{dehyd}
\end{aligned} \tag{2.77}$$

Momentum conservation

By neglecting inertial forces:

$$\nabla\cdot\boldsymbol{\sigma}^{Total} + \rho\,\mathbf{g} = \mathbf{0} \tag{2.78}$$

where $\mathbf{g}$ is the gravitational acceleration and the total density is:

$$\rho = (1-\phi)\rho_s + \phi\left(S_l \rho_l + (1-S_l)\rho_g\right) \tag{2.79}$$

The total stress $\boldsymbol{\sigma}^{Total}$ is related to the effective stress via Bishop's relation (Eq. (**??**)), which in turn depends on the primary variables p_g and p_c.

The governing equations presented in this chapter — dry air conservation, water species conservation, energy conservation, and momentum conservation — together with the constitutive laws, form a complete THM model for concrete exposed to fire. These equations are continuous partial differential equations defined on the domain Ω. Their numerical solution requires spatial and temporal discretization, which is the subject of the following Chapter 3.

2.5.2 Spalling mechanisms

Spalling is defined as the violent or non-violent detachment of layers or pieces of concrete from the surface of a structural element during or after exposure to rapidly rising temperatures. It may lead to premature structural failure through reinforcement exposure and cross-section reduction. Several catastrophic tunnel fires (Channel Tunnel 1996, Mont-Blanc 1999, St. Gotthard 2001) have illustrated the severity of this phenomenon [3].

Despite extensive experimental research, the mechanisms behind spalling are not yet fully understood and no general consensus on a single theory has been reached. The two principal mechanisms most broadly accepted are presented below, followed by their combined action.

Thermal gradient mechanism

[REVIEW] This section is currently under development and will be refined as the study progresses.

This mechanism is related to the thermo-mechanical processes. When one face of a concrete member is exposed to fire, a steep temperature gradient develops through the thickness. The resulting differential thermal dilation produces (see Figure **??**):

1. **Compressive stresses in the heated cover:** The surface layers tend to expand but are restrained by the cooler interior, generating significant compressive stresses parallel to the heated surface.

2. **Tensile stresses in the interior:** The restraint of the expanding surface layers induces tensile stresses in deeper layers of the material.

3. **Fracture and energy release:** When the tensile stress exceeds the (temperature-degraded) tensile strength $f_t(T)$, cracks develop parallel to the surface. The sudden, unstable release of the elastic energy stored in the compressed surface layers can drive the explosive ejection of concrete fragments.

In terms of the THM model, this mechanism involves the energy equation (2.77) (which determines the temperature field), the thermal dilation strain (Eq. (2.69)), and the Mazars damage model which captures the tensile cracking.

High performance concrete (HPC) is more prone to thermal gradient-induced spalling because it is far more *brittle* than normal strength concrete: the higher elastic energy stored before failure leads to a more violent release.

Pore pressure mechanism (moisture clog)

[REVIEW] This section is currently under development and will be refined as the study progresses.

This mechanism is related to the thermo-hydral processes. The sequence of events is (see Figure **??**):

1. **Formation of a dry zone:** As the surface is heated, free water evaporates and chemically bound water is released as vapor through dehydration. A dry zone forms near the heated face.

2. **Vapor migration:** Due to pressure gradients, part of the vapor is evacuated toward the heated face, while part is driven toward the cooler interior.

3. **Condensation and moisture clog:** The inward-migrating vapor condenses when it reaches regions where the thermodynamic conditions are satisfied ($T < T_{sat}$). This process continues until a quasi-saturated zone is formed, known as the *moisture clog*, which acts as an impermeable barrier preventing further vapor migration toward the cold side.

4. **Pressure build-up:** Behind the moisture clog, the gas pressure p_g increases sharply as continued evaporation feeds vapor into a region from which it cannot escape. In addition, the thermal dilation of liquid water in saturated pores also contributes to pressure build-up.

5. **Tensile failure:** When the pore pressure exceeds the tensile strength of the material, cracks develop parallel to the surface, potentially leading to explosive spalling.

In terms of the THM model, this mechanism involves the water species conservation equation (2.76) (which determines the pressure field), the permeability law Equation (2.39)), and the effective stress (Eq. (**??**)) through which pore pressure loads the solid skeleton.

The key material parameters governing this mechanism are the *intrinsic permeability* K_0 and the *initial moisture content*. Low permeability hinders vapor evacuation, while high moisture content provides more water to be evaporated, both promoting higher pore pressures.

Combined mechanism and HPC vulnerability

[REVIEW] This section is currently under development and will be refined as the study progresses.

In practice, explosive spalling generally occurs under the *combined action* of pore pressures and thermal stresses. The spalling condition can be expressed schematically as:

$$\sigma_{thermal}(T) + \sigma_{pore}(p_g, p_c, S_l) \geq f_t(T) \tag{2.80}$$

where $\sigma_{thermal}$ represents the thermal stress from differential dilation, σ_{pore} represents the stress from pore pressure loading (via effective stress), and $f_t(T)$ is the temperature-degraded tensile strength. When the sum of stresses exceeds the tensile strength, cracks develop parallel to the heated surface, often leading to violent failure of the fire-exposed region.

This combined criterion highlights why the complete THM model is needed: neither a purely thermal analysis (T only) nor a purely hydral analysis (p_g, p_c only) can capture the full picture.

[REVIEW]

High Performance Concrete (HPC) is particularly vulnerable to spalling for several compounding reasons, all of which are captured by the THM model:

Role of damage-permeability coupling

[REVIEW] This section is currently under development and will be refined as the study progresses.

The competition between pore pressure build-up and damage-induced permeability increase (Eq. (2.39)) plays a decisive role in determining whether spalling occurs:

- **Scenario 1 — No spalling:** If microcracking develops early (e.g., from thermal

Table 2.2: Factors contributing to HPC vulnerability to spalling

HPC property	**Model parameter**	**Effect on spalling**
Low water-to-cement ratio	Low intrinsic permeability K_0	Hinders vapor evacuation $\rightarrow$ higher p_g
Dense pore structure	Low porosity ϕ_0	Less storage volume for vapor $\rightarrow$ rapid pressure build-up
High compressive strength	Higher E_0	More elastic energy stored $\rightarrow$ more violent release
Higher brittleness	Lower G_f/f_t ratio	Less energy dissipation before fracture $\rightarrow$ sudden failure
Higher initial saturation	Higher $S_{l,0}$	More water available for evaporation $\rightarrow$ higher p_g

stresses), the damage variable D increases, causing an increase in permeability via the factor $10^{A_d \cdot d}$. This creates additional escape paths for vapor, *reducing* pore pressure before it can reach the critical level. This self-regulating mechanism is more effective in normal-strength concrete, which has higher initial permeability and cracks more readily.

- **Scenario 2 — Spalling:** In HPC, the low initial permeability causes rapid pressure build-up, while the high tensile strength delays cracking. By the time the first cracks form, the pore pressure may already exceed the tensile strength, leading to sudden, explosive failure before permeability can increase sufficiently.

This explains the paradox that *stronger* concrete is often *more* vulnerable to spalling than weaker concrete.

Factors influencing spalling

In addition to the material properties discussed above, the main parameters promoting spalling, as identified by experimental research, include:

- **Moisture content:** Higher initial moisture provides more water for evaporation and therefore higher peak pore pressures;
- **Heating rate:** Faster heating creates steeper thermal gradients and allows less time for pressure relief through transport;
- **External loading:** Compressive loading superimposes additional stresses to the thermally induced ones, reducing the margin before tensile failure;

- **Concrete strength and permeability:** As discussed above, higher strength and lower permeability increase vulnerability;
- **Aggregate type:** Mineralogical character of aggregates affects thermal expansion compatibility and crack initiation;
- **Geometry and element size:** Thinner sections and edges are more prone to spalling.

The THM model thus provides a unified framework to explain, quantify, and predict spalling by computing the spatial and temporal evolution of the thermal, hydric, and mechanical fields and their interactions.

Chapter 3

Numerical implementation

[REVIEW] This chapter is currently empty and is intended to outline the proposed structure of the thesis.

This chapter describes the numerical solution of the THM governing equations developed in Chapter 2. The implementation proceeds incrementally: the thermal, hydric, and mechanical models are first solved separately, then coupled to form the full THM system. All implementations are carried out in the finite element code Cast3M.

3.1 Discretization

3.1.1 Spatial discretization

[REVIEW] This section is currently under development and will be refined as the study progresses.

Applying the standard Galerkin finite element method to the three TH conservation equations (Equations (2.75) to (2.77)) yields the semi-discrete system [3]:

$$\mathbf{C}\,\dot{\mathbf{Y}} + \mathbf{K}\,\mathbf{Y} = \mathbf{f} \tag{3.1}$$

where $\mathbf{Y} = \{\bar{p}_g, \bar{p}_c, \bar{T}\}$ is the vector of nodal TH unknowns. The operators $\mathbf{C}$ (capacity), $\mathbf{K}$ (conductivity), and $\mathbf{f}$ (forcing) are 3×3 block matrices:

$$\mathbf{K} = \begin{pmatrix} K_{gg} & K_{gc} & K_{gt} \\ K_{cg} & K_{cc} & K_{ct} \\ K_{tg} & K_{tc} & K_{tt} \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} C_{gg} & C_{gc} & C_{gt} \\ 0 & C_{cc} & C_{ct} \\ 0 & C_{tc} & C_{tt} \end{pmatrix}, \quad \mathbf{f} = \begin{pmatrix} f_g \\ f_c \\ f_t \end{pmatrix} \tag{3.2}$$

The off-diagonal entries K_{ij}, C_{ij} $(i \neq j)$ encode the coupling between the thermal and hydric fields. Detailed expressions of each block are given in [3], Appendix A.

The mechanical problem (Equation (2.78)) is solved separately in a staggered manner: at each time step, the TH solution provides T, p_g, p_c,which define the thermal strain (Equation (2.69)), shrinkage strain (Equation (2.71)), and pore pressure loading (Equation (2.73)). The equilibrium equation:

$$\mathbf{K}_M\,\mathbf{u} = \mathbf{F}(T,\, p_g,\, p_c) \tag{3.3}$$

is then solved for the displacement field $\mathbf{u}$, where $\mathbf{K}_M$ is the mechanical stiffness matrix and $\mathbf{F}$ is the load vector incorporating thermo-hydric contributions.

3.1.2 Temporal discretization

[REVIEW] This section is currently under development and will be refined as the study progresses.

The time integration of Equation (3.1) is carried out using the Wilson-θ method [3], which evaluates the equation at an intermediate time $t_{n+\theta} = t_n + \theta\,\Delta t$ with $\theta \in [0, 1]$. This transforms the semi-discrete system into a fully discrete algebraic system at each time step:

$$\tilde{\mathbf{K}}\,\Delta\mathbf{Y}^{n+1} + \mathbf{K}\,\mathbf{Y}^{n} = \mathbf{f}^{n+\theta} \tag{3.4}$$

where:

$$\Delta\mathbf{Y}^{n+1} = \mathbf{Y}^{n+1} - \mathbf{Y}^{n}, \qquad \tilde{\mathbf{K}} = \frac{\mathbf{C}}{\Delta t} + \theta\,\mathbf{K} \tag{3.5}$$

The parameter θ controls the scheme: $\theta = 0$ gives the explicit forward Euler method, $\theta = 1$ the fully implicit backward Euler, and $\theta = 0.5$ the Crank-Nicolson scheme. In practice, $\theta = 1$ is used for unconditional stability.

Since the capacity and conductivity matrices depend nonlinearly on the unknowns ($\mathbf{C} = \mathbf{C}(\mathbf{Y})$, $\mathbf{K} = \mathbf{K}(\mathbf{Y})$), Equation (3.4) is solved iteratively using the Newton-Raphson method within each time step.

3.1.3 Solution strategy

[REVIEW] This section is currently under development and will be refined as the study progresses.

The TH subsystem (Equation (3.4)) is solved *monolithically*: all three unknowns p_g, p_c, T are updated simultaneously within a single Newton-Raphson loop at each time step.

The mechanical problem (Equation (3.3)) is solved in a *staggered* manner: at each time step, the converged TH solution provides T, p_g, p_c, which are then used to assemble $\mathbf{F}$

and solve for $\mathbf{u}$. The damage variable $\mathcal{D}$ computed from the Mazars model feeds back to the intrinsic permeability via ??, creating a weak M→H coupling that is updated at the next time step.

3.2 Modeling strategy and model parameters

3.2.1 Modeling strategy

This section is currently under development and will be refined as the study progresses.

3.2.2 Geometry

This section is currently under development and will be refined as the study progresses.

```
OPTI ECHO 0;
OPTI 'DIME' 2 'ELEM' 'QUA4' ;

* 1) GEOMETRY AND MESH DEFINITION

*GEOMETRY DEFINITION
LONG    = 3.;
HAUT    = 0.5;

*NUMBER OF MESH SEGMENT
NLONG   = 35;
NHAUT   = 5;

*POINTS CREATION
PA      = 0. 0.;
PB      = LONG 0.;
PC      = LONG HAUT;
PD      = 0. HAUT;
TRAC (PA ET PB ET PC ET PD);

*MESH LINES CREATION
LAB     = DROI NLONG PA PB;
LBC     = DROI NHAUT PB PC;
LCD     = DROI NLONG PC PD;
LDA     = DROI NHAUT PD PA;

```

```
*MESH SURFACE CREATION
MAIL1    = DALL LAB LBC LCD LDA;
TRAC MAIL1;
```

3.2.3 Material properties

[REVIEW]

High-strength concrete with $f_{c,28} = 77$ MPa is adopted, and 18 key parameters are selected for sensitivity analysis [6].

3.2.4 Boundary conditions and loadings

This section is currently under development and will be refined as the study progresses.

3.3 Implementation of the thermal model

3.3.1 Stationary linear thermal analysis

This section is currently under development and will be refined as the study progresses.

3.3.2 Transient linear and non-linear thermal analysis

This section is currently under development and will be refined as the study progresses.

3.4 Implementation of the hydric model

3.4.1 Drying

This section is currently under development and will be refined as the study progresses.

3.4.2 Shrinkage

This section is currently under development and will be refined as the study progresses.

3.5 Implementation of the mechanical model

3.5.1 Elastic linear mechanical model

This section is currently under development and will be refined as the study progresses.

3.5.2 Mazars model

This section is currently under development and will be refined as the study progresses.

3.6 Implementation of the TH coupled model

This section is currently under development and will be refined as the study progresses.

3.7 Implementation of the THM coupled model

3.7.1 TH coupling strategy

This section is currently under development and will be refined as the study progresses.

3.7.2 TM coupling strategy

This section is currently under development and will be refined as the study progresses.

3.7.3 THM coupling strategy

This section is currently under development and will be refined as the study progresses.

3.8 Numerical results

This section is currently under development and will be refined as the study progresses.

Chapter 4

Experimental data and Model validation

[REVIEW] This chapter is currently empty and is intended to outline the proposed structure of the thesis.

4.1 Experimental data

4.1.1 Experimental tests and material properties

This section is currently under development and will be refined as the study progresses.

4.1.2 Thermal, hydraulic and mechanic loadings

This section is currently under development and will be refined as the study progresses.

4.1.3 Measured data

This section is currently under development and will be refined as the study progresses.

4.2 Validation

4.2.1 Comparison between numerical and experimental results

This section is currently under development and will be refined as the study progresses.

4.2.2 Discussions on model performance

This section is currently under development and will be refined as the study progresses.

Chapter 5

Probabilistic analysis of the THM model

[REVIEW] This chapter is currently a draft version and will undergo further review and revision.

5.1 Sources of uncertainty and concrete heterogeneity

This section follows the foundational framework presented in [1].

[REVIEW]

Main ideas:

- Concrete is not uniform, so its properties vary from place to place.
- Material parameters (e.g., stiffness and permeability) are uncertain.
- Measurements and model assumptions also add uncertainty.

5.1.1 Motivation for probabilistic assessment

In civil engineering, a structure or mechanical system is considered safe when it remains fit for duty for the duration of its service life, without causing damage to itself or its environment. Traditionally, deterministic models have been used to evaluate structural safety. However, the inevitable gap between a real physical system and a simplified numerical model, along with the unforeseeable nature of future loads and environmental conditions, inherently leads to uncertainty.

5.1.2 Classification of uncertainties

Predicting the exact lifetime or mechanical response of a construction project is impossible without accounting for multiple sources of uncertainty. According to reliability theory, these uncertainties can be broadly classified into two main categories: Aleatoric and Epistemic.

5.1.3 Concrete heterogeneity and material variability

A critical source of aleatoric uncertainty in civil engineering is the inherent heterogeneity of construction materials, such as concrete and geomaterials.

- **Historical and environmental dependence:** Concrete is not perfectly uniform. Its physical and mechanical properties depend heavily on its entire formation history, the conditions during casting, internal chemical reactions (like alkali-silica reactions), and long-term degradation processes caused by the surrounding environment.
- **Continuous and discrete spatial variations:** Because of this historical dependence, material parameters vary significantly from place to place. This spatial heterogeneity manifests in two ways:
  1. *Continuous variations:* Gradual spatial fluctuations in properties such as stiffness, Young's modulus, Poisson's ratio, density, or friction.
  2. *Discrete variations:* The localized appearances of defects such as micro-cracks, water inlets, networks of capillaries, or weak points within the structure.

5.2 Stochastic finite element framework (Monte Carlo)

[REVIEW]

Main ideas:

- Treat uncertain material parameters as random variables.
- Generate many random samples.
- Run the THM model for each sample.
- Collect the outputs and analyze variability and sensitivity.

5.2.1 Random Field theory and Spatial variability

[REVIEW] This section is currently under development and will be refined as the study progresses.

Random fields

Spatial variability of material properties can be described using random field theory, in which a parameter is modeled as a stochastic function defined over the spatial domain. In this representation, the value of the property varies from point to point according to a prescribed probability distribution and spatial correlation structure. Random fields therefore provide a convenient framework to represent the heterogeneous nature of materials such as concrete.

Discretized random field

For numerical simulations, the continuous random field must be discretized in order to be implemented within a finite element model. This discretization assigns random values of the considered property to the elements or integration points of the mesh. In this way, the spatial variability of the material parameters can be incorporated directly into the computational model.

Spatial correlation length (λ)

[REVIEW] This section requires additional references to support the statements presented.

A fundamental parameter in the generation of random fields is the spatial correlation length, often denoted as λ (or b).

Depending on the assigned value of λ:

- A very small correlation length ($\lambda \to 0$) implies that the material properties fluctuate rapidly and erratically over short distances
- A large correlation length ($\lambda \to \infty$) implies a slow spatial variation, moving towards a uniformly homogeneous material

[REVIEW]

In the context of this study, the choice of the correlation length is closely related to the maximum aggregate size of the concrete mixture.

5.2.2 Transformation of Lognormal to Gaussian distributionn

In practical finite element implementations using the Cast3M software, material parameters like the Young's modulus (E) are often assigned a lognormal random field to ensure the generated stiffness values remain strictly positive. However, the built-in random field operator (`ALEA`) generates Gaussian (normal) distributions, necessitating a mathematical transformation.

[REVIEW]

[REVIEW] This section requires additional references to support the statements presented.

Assuming the actual mean of the material's Young's modulus is μ_E and its standard deviation is σ_E. The fundamental relationship between these two variables is expressed through the exponential transformation:

$$E = \exp(G) \iff G = \ln(E)$$

From statistical properties, we can establish a system of two equations relating the moments of E (μ_E, σ_E) and G (μ_G, σ_G) as follows:

$$\begin{cases} \mu_E = \exp\left(\mu_G + \dfrac{\sigma_G^2}{2}\right) \\ \sigma_E^2 = [\exp(\sigma_G^2) - 1] \exp(2\mu_G + \sigma_G^2) \end{cases} \tag{5.1}$$

To find the input parameters μ_G and σ_G for the random number generation algorithm, we need to solve the above system of equations.

1. Solving for σ_G^2

By squaring the first equation, we get:

$$\mu_E^2 = \exp(2\mu_G + \sigma_G^2)$$

Substituting this result into the second equation yields:

$$\sigma_E^2 = \left[\exp(\sigma_G^2) - 1\right] \mu_E^2$$

Rearranging the terms, we obtain:

$$\exp(\sigma_G^2) = \frac{\sigma_E^2}{\mu_E^2} + 1$$

Taking the natural logarithm of both sides gives the variance of the Gaussian distribution:

$$\sigma_G^2 = \ln\left(1 + \frac{\sigma_E^2}{\mu_E^2}\right)$$

2. Solving for μ_G

Using the first equation and isolating $\exp(\mu_G)$, we have:

$$\mu_E = \exp(\mu_G) \cdot \exp\left(\frac{\sigma_G^2}{2}\right)$$

Substituting $\sigma_G^2 = \ln\left(1 + \frac{\sigma_E^2}{\mu_E^2}\right)$ into the exponential term, we get:

$$\mu_E = \exp(\mu_G) \cdot \exp\left(\ln\left(\sqrt{1 + \frac{\sigma_E^2}{\mu_E^2}}\right)\right) = \exp(\mu_G) \cdot \sqrt{1 + \frac{\sigma_E^2}{\mu_E^2}}$$

Rearranging to solve for μ_G:

$$\exp(\mu_G) = \frac{\mu_E}{\sqrt{1 + \frac{\sigma_E^2}{\mu_E^2}}} \implies \mu_G = \ln\left(\frac{\mu_E}{\sqrt{1 + \frac{\sigma_E^2}{\mu_E^2}}}\right)$$

Summary of calculation steps in the numerical algorithm

Based on the exact solutions derived above, the implementation follows these sequential steps:

1. The coefficient of variation (cov_E) of the desired lognormal distribution is calculated:

$$cov_E = \frac{\sigma_E}{\mu_E}$$

2. The mean of the equivalent Gaussian distribution (μ_G):

$$\mu_G = \ln\left(\frac{\mu_E}{\sqrt{1 + cov_E^2}}\right)$$

3. The variance of the equivalent Gaussian distribution (σ_G^2):

$$\sigma_G^2 = \ln(1 + cov_E^2)$$

4. The standard deviation of the Gaussian distribution (σ_G):

$$\sigma_G = \sqrt{\ln(1 + cov_E^2)}$$

5. A standard Gaussian random field (`GFIELD`) is then generated using the `ALEA` operator with the computed Gaussian mean μ_G and standard deviation σ_G.

6. Finally, this Gaussian field is transformed back into the actual lognormal distribution (`EFIELD`) using the exponential function. A scale factor of 10^9 is multiplied to convert the physical unit from GPa to Pa for the mechanical solver:

$$EFIELD = \exp(GFIELD) \times 10^9$$

5.2.3 Integration of spatial variability into the THM model

This section is currently under development and will be refined as the study progresses.

5.2.4 Monte Carlo simulation framework

After successfully implementing the log-normal random fields to represent the concrete's spatial heterogeneity, the next step in the probabilistic assessment is to perform multiple independent realizations. This is achieved through a Monte Carlo simulation framework.

In the finite element code Cast3M, this iterative process is seamlessly governed by the `REPETER` operator.

Simulation loop and dynamic material updating

This section is currently under development and will be refined as the study progresses

Data extraction and post-processing

This section is currently under development and will be refined as the study progresses

5.3 Gaussian Quadrature framework

This section is currently under development and will be refined as the study progresses.

5.3.1 Principle of Gaussian Quadrature

[REVIEW]

Main ideas as follows:

From integration to weighted sum

Orthogonal polynomials and weighting functions

Abscissas and weights construction

5.3.2 Statistical moment estimation

[REVIEW]

Main ideas as follows:

Mean, variance, skewness, kurtosis from quadrature

Variance decomposition per variable

5.3.3 PDF, CDF and reliability assessment

[REVIEW]

Main ideas as follows:

Approximation methods (Pearson, Winterstein, Exp-Polynomial)

Probability of failure and reliability index

5.3.4 Computational efficiency

[REVIEW]

Main ideas as follows:

Cost comparison with Monte Carlo

Quick Quadrature for high-dimensional problems

Data extraction: moments computed directly from weighted sum

Limitations: curse of dimensionality

5.4 Sensitivity analysis

[REVIEW]

Main ideas:

- Sensitivity indicators: Young modulus...
- Change one input parameter at a time (or groups of parameters).
- Measure how much the outputs change.
- Rank the parameters from most to least influential.

5.4.1 Characterization of input parameters

This section is currently under development and will be refined as the study progresses.

Characterize the chosen input parameters in terms of domain variations and coefficient of variation.

[REVIEW]
Main ideas as follows:

Chosen Random variables

Correlated vs uncorrelated parameters

Physical correlations

5.4.2 Sensitivity indicators and ranking methods

This section is currently under development and will be refined as the study progresses.

Pearson correlation, Sobol indices (if variance-based), or Sensitivity derivatives (OAT - One-At-a-Time)

[REVIEW]
Main ideas as follows:

Variance decomposition

Uncorrelated parameters

Correlated parameters

Quadrature: one-variable-at-a-time moments

Monte Carlo: Sobol indices and Pearson correlation

5.4.3 Identification of dominant parameters

This section is currently under development and will be refined as the study progresses.

[REVIEW]
Main ideas as follows:

Ranking from most to least influential

Comparison: Quadrature ranking vs Monte Carlo ranking

Physical interpretation of dominant parameters

5.5 Reliability analysis

[REVIEW]

Main ideas as follows:

5.5.1 Probability of failure and reliability index

This section is currently under development and will be refined as the study progresses.

5.5.2 Probability of failure from moment-based PDF

This section is currently under development and will be refined as the study progresses.

5.5.3 Probability of failure from Monte Carlo sampling

This section is currently under development and will be refined as the study progresses.

5.5.4 Effect of spatial heterogeneity on structural safety

This section is currently under development and will be refined as the study progresses.

Chapter 6

Conclusions and Perspectives

[REVIEW] This chapter is currently empty and is intended to outline the proposed structure of the thesis.

6.1 Main findings

This section is currently under development and will be refined as the study progresses.

6.2 Model limitations and possible improvements

This section is currently under development and will be refined as the study progresses.

6.3 Perspectives for future research

This section is currently under development and will be refined as the study progresses.

List of Figures

List of Tables

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